Théorie des Langages

4 Sujets à traiter

→ Sujet 1:

Détermination du langage accepté par un automate (par son ER)

→ Sujet 2 :

Rendre complet un AEF

Rendre non ambigu un AEF

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Rendre déterministe un AFN sans ε-transitions

→ Sujet 3 :

Rendre déterministe un AFN avec ε-transition

→ Sujet 4 :

Minimiser un AFD

Sujet 1

Automate → ? → ER

1) Langage accepté par un automate
2) Langage accepté à partir d'un état
3) Les équations d'~~un langage~~ Automate
4) Lemme d'Arden

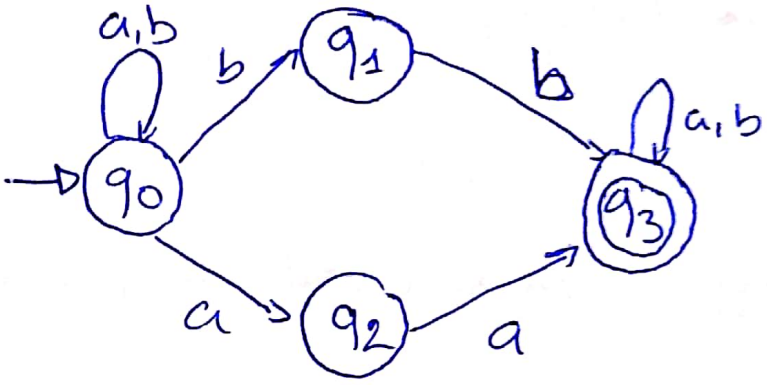
$M = (Q, \Sigma, \delta, S = q_0, F)$

$\mathcal{L}(M) = \{ w \in \Sigma^* ; (q_0, w) \vdash^* (q_f, \varepsilon) \text{ avec } q_f \in F \}$

$\mathcal{L}(M) = \mathcal{L}(q_0) = L_{q_0}$

$\mathcal{L}(q) = L_q = \{ w \in \Sigma^* ; (q, w) \vdash^* (q_f, \varepsilon) \text{ avec } q_f \in F \}$

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Ecrire L_{q_0} en fonction L_{q_1} et L_{q_2}

$$L_{q_0} = \{a\} \cup \{b\} L_{q_0} \cup \{b\} L_{q_1} \cup \{a\} L_{q_2}$$

ou bien

$$(a \cup b) L_{q_0} \cup b L_{q_1} \cup a L_{q_2}$$

$A = (Q, \Sigma, \delta, S = q_0, F)$

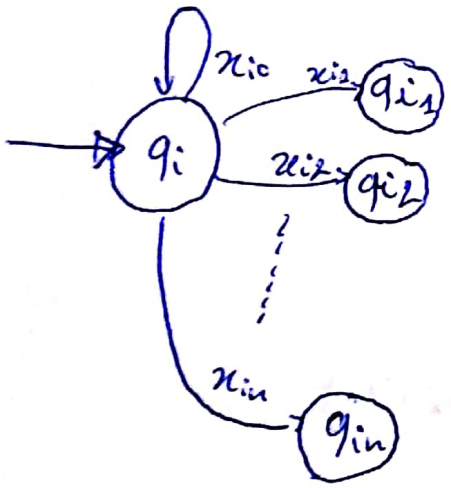
Les équations de l'automate

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Les expressions des Langages des différents états de l'automate

Ex: $Q = \{q_0, q_1, q_2\}$

Equations : $\begin{cases} L_{q_0} = \dots \\ L_{q_1} = \dots \\ L_{q_2} = \dots \end{cases}$

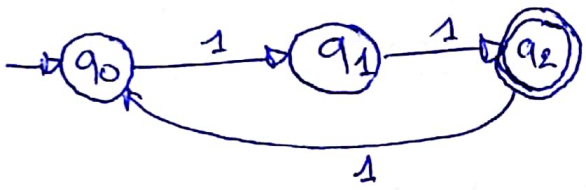


$$L_{q_i} = (x_{i0} L_{q_i} \cup x_{i1} L_{q_{i1}} \cup \dots \cup x_{in} L_{q_{in}}) \cup \alpha$$

$$\alpha = \begin{cases} \{\varepsilon\} & \text{si } q_i \text{ est final} \\ \emptyset & \text{si } q_i \text{ n'est pas final} \end{cases}$$

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A Automate



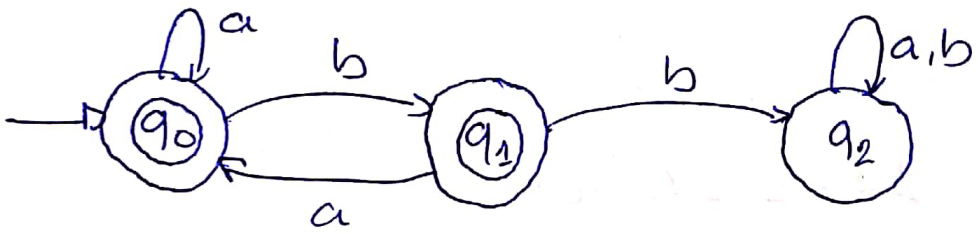
Les expressions:

$L_{q_0} = 1 . L_{q_1}$

$L_{q_1} = 1 L_{q_2}$

$L_{q_2} = 1 L_{q_0} \cup \varepsilon$

Automate M1



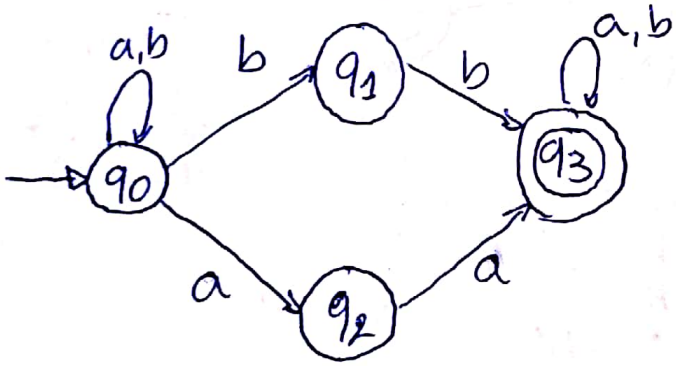
Les expressions:

$L_{q_0} = a L_{q_0} \cup b L_{q_1} \cup \varepsilon$ (car q_0 : final)

$L_{q_1} = a L_{q_0} \cup b L_{q_2} \cup \varepsilon$ (car q_1 : final)

$L_{q_2} = (a \cup b) L_{q_2}$

Automate M3



Les expressions:

$L_{q_0} = (a \cup b) L_{q_0} \cup b L_{q_1} \cup a L_{q_2}$

$L_{q_1} = b L_{q_3}$

$L_{q_2} = a L_{q_3}$

$L_{q_3} = (a \cup b) L_{q_3} \cup \varepsilon$

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Lemme d'Arden:

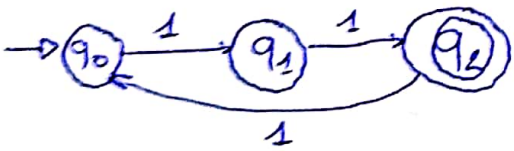
$L = AL \cup B$ ou $L = LA \cup B$

Si $\varepsilon \notin A$: Cette équation a une solution unique.

A^*B ou BA^+

Exemple: Déterminez le langage accepté/reconnu par l'automate.

Equations des langages de l'automate:



$L_0 = 1L_1$ (1)

$L_1 = 1.L_2$ (2)

$L_2 = 1.L_0 \cup \varepsilon$ (3)

À partir de (2) et (3): $L_1 = 1(1L_0 \cup \varepsilon)$

$L_1 = 11L_0 \cup 1$ (4)

À partir de (1) et (4): $L_0 = 1(11L_0 \cup 1)$

$L_0 = 111L_0 \cup 11$

Solution: $\boxed{L_0 = (111)^*11}$

Exemple: $\Sigma = \{a, b\}$

1- Automate qui accepte le langage $(a \cup b)^*$

a-



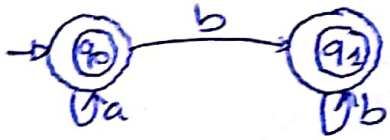
b- $L_0 = (a \cup b)L_0 \cup \varepsilon$

$\varepsilon \notin (a \cup b)$

$L_0 = (a \cup b)^*\varepsilon = (a \cup b)^*$

Automate qui accepte le langage a^*b^*

2- a-



b-

$L_0 = aL_0 \cup bL_1 \cup \varepsilon$

$L_1 = bL_1 \cup \varepsilon$ (1)

On applique Arden à L_1: $(\varepsilon \notin \{b\})$

$L_1 = b^*\varepsilon = b^*$

On remplace $L_1 = b^*$ dans (1)

$L_0 = aL_0 \cup bb^* \cup \varepsilon = aL_0 \cup (bb^* \cup \varepsilon)$

$L_0 = aL_0 \cup b^*$ $(car\ (bb^* \cup \varepsilon) = b^*)$ $(bb^* = b^+;\ b^+ \cup \varepsilon = b^*)$ (4)

On applique Arden:

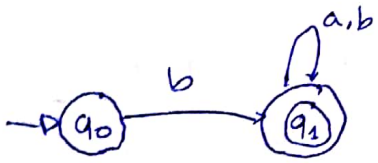
$\boxed{L_0 = a^*b^*}$

TZA

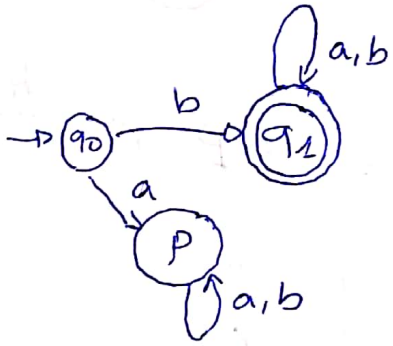
Sujet 2

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AFN ⟹ AFD

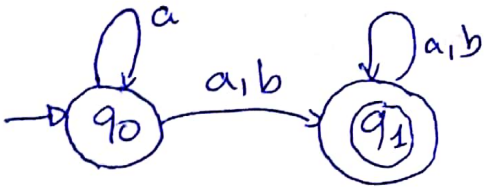


$\delta(q_0, a) = \phi$



$\delta(q_0, a) = \{P\}$

Désambiguiser un automate :



q_i \ Σ	a	b
→ q_0	$\{q_0, q_1\}$	$\{q_1\}$
* q_1	$\{q_1\}$	$\{q_1\}$

Etape 1 : $\{0\}$ (Etat initial)

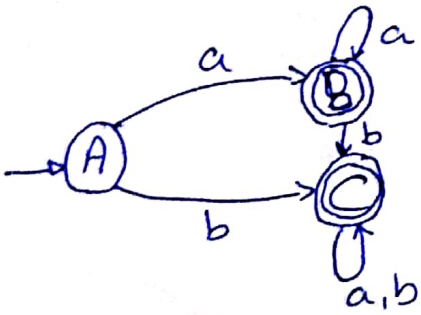
Etape 2 : Création de nouveau état

	a	b
$\{q_0\}_A$	$\{q_0, q_1\}_B$	$\{q_1\}_C$
$\{q_0, q_1\}_B$	$\{q_0, q_1\}_B$	$\{q_1\}_C$
$\{q_1\}_C$	$\{q_1\}_C$	$\{q_1\}_C$

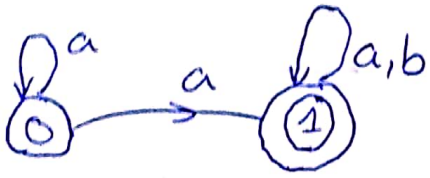
Etape 3 : Renommage d'état

δ	a	b
→ A	B	C
* B	B	C
* C	C	C

Etape 4 : Construction de l'automate.



AFN



δ	a	b
{0}A	{0,1}B	∅ P(puit)
{0,1}B	{0,1}B	{1}C
{1}C	{1}C	{1}C

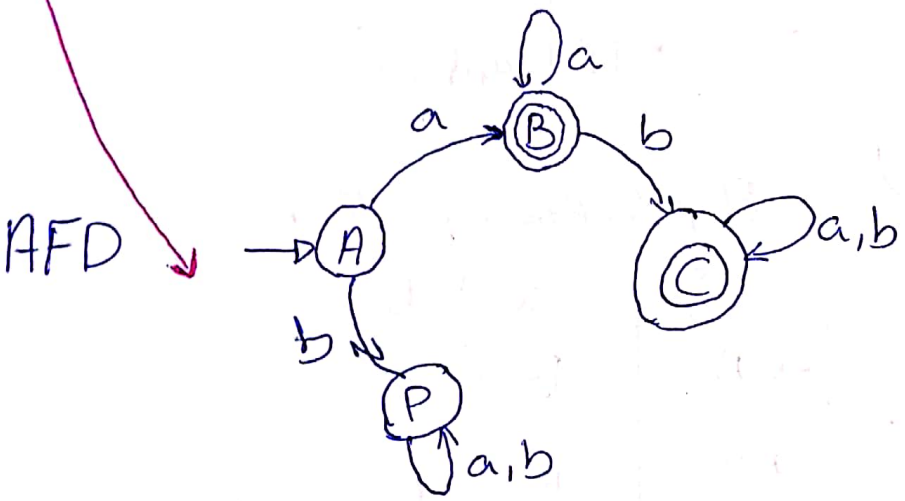
Déterminisation

AFD →

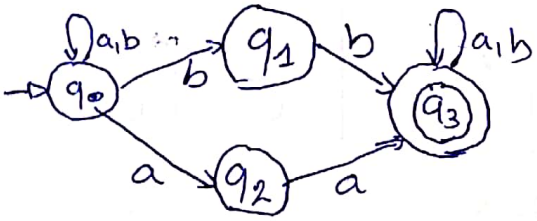
δ	a	b
→ A	B	P
* B	B	C
* C	C	C

Construction de l'automate

AFD



Automate M3

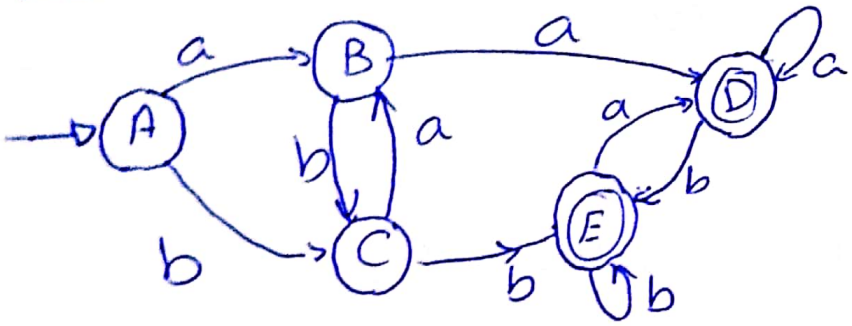


δ	a	b
{q0}A	{q0,q2}B	{q0,q1}C
{q0,q2}B	{q0,q2,q3}D	{q0,q1}C
{q0,q1}C	{q0,q2}B	{q0,q1,q3}E
{q0,q2,q3}D	{q0,q2,q3}D	{q0,q1,q3}E
{q0,q1,q3}E	{q0,q2,q3}D	{q0,q1,q3}E

⟹

δ	a	b
→ A	B	C
B	D	C
C	B	E
* D	D	E
* E	D	E

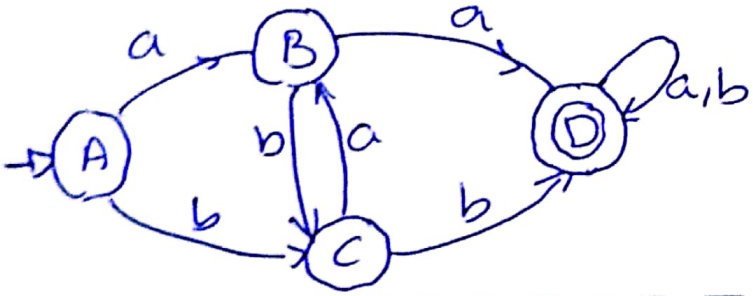
Construction de l'automate :



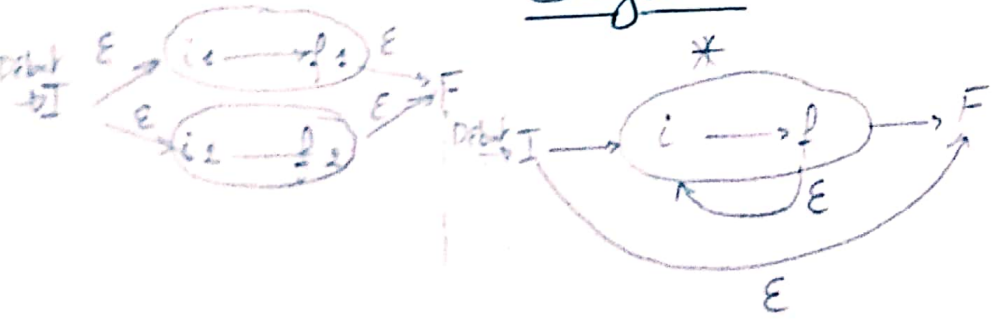
Extra : $L_D = aL_D \cup bL_E \cup \varepsilon$

$L_E = aL_D \cup bL_E \cup \varepsilon$

On peut simplifier l'automate puisque $L_D = L_E$



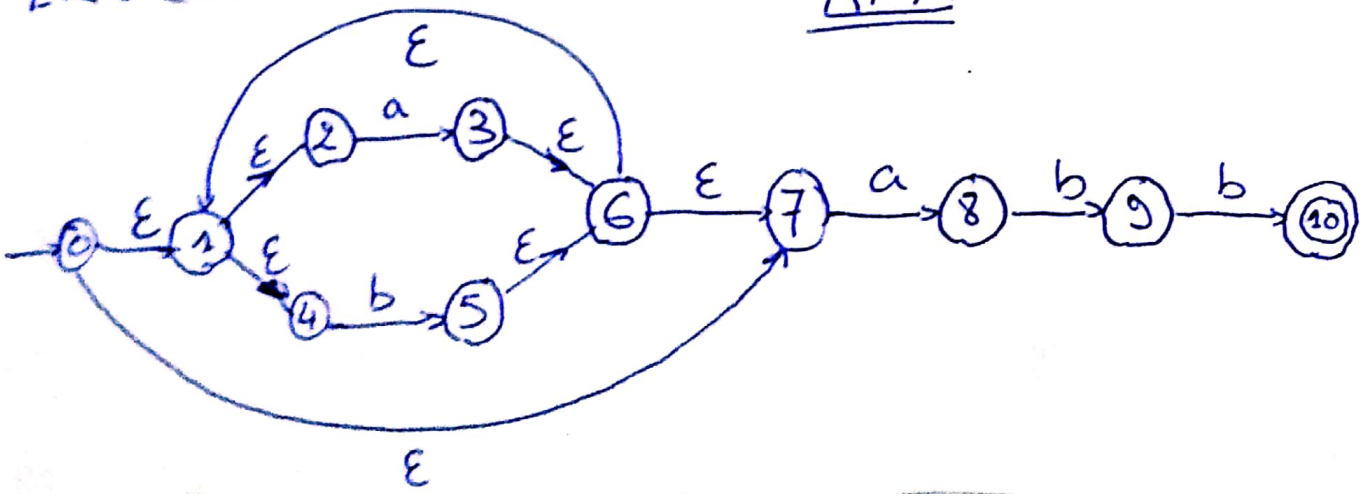
Sujet 3



Exemple :

ER = (a∪b)*.aab

AFN



+ E-fermeture({0}) = {0, 1, 2, 4, 7} = A l'ensemble des états qu'on peut passer par E-transitions à partir de 0

On cherche la trans. (X, w) = Y puis E-fermeture(Y)

A = {0, 1, 2, 4, 7}

E-fermeture(transition(A, a)) = E-fermeture({3, 8})

= {3, 8, 6, 7, 1, 2, 4}

E-fermeture({3, 8}) = {1, 2, 3, 4, 6, 7, 8} = B

8 n'a pas des E-transitions